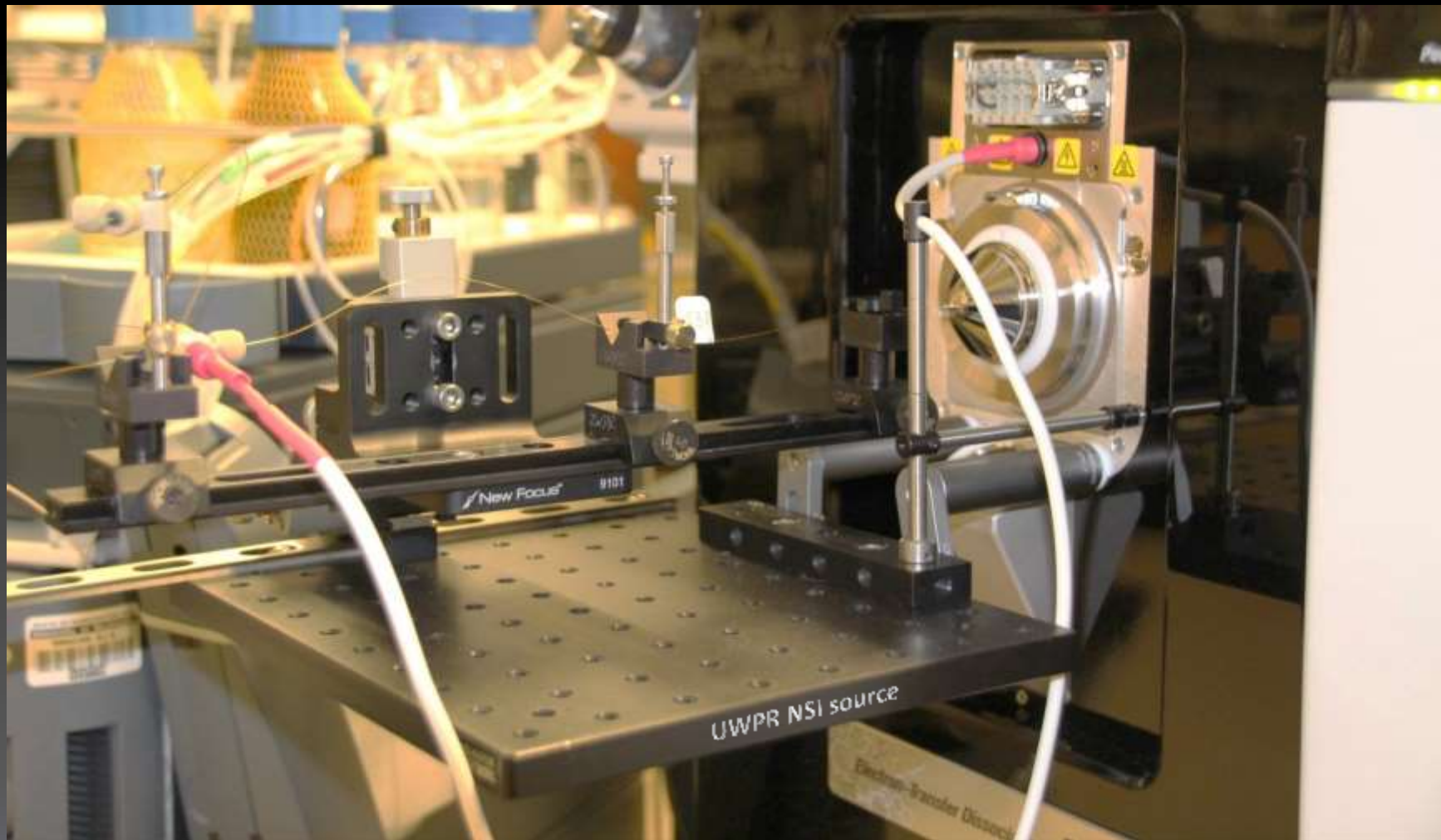


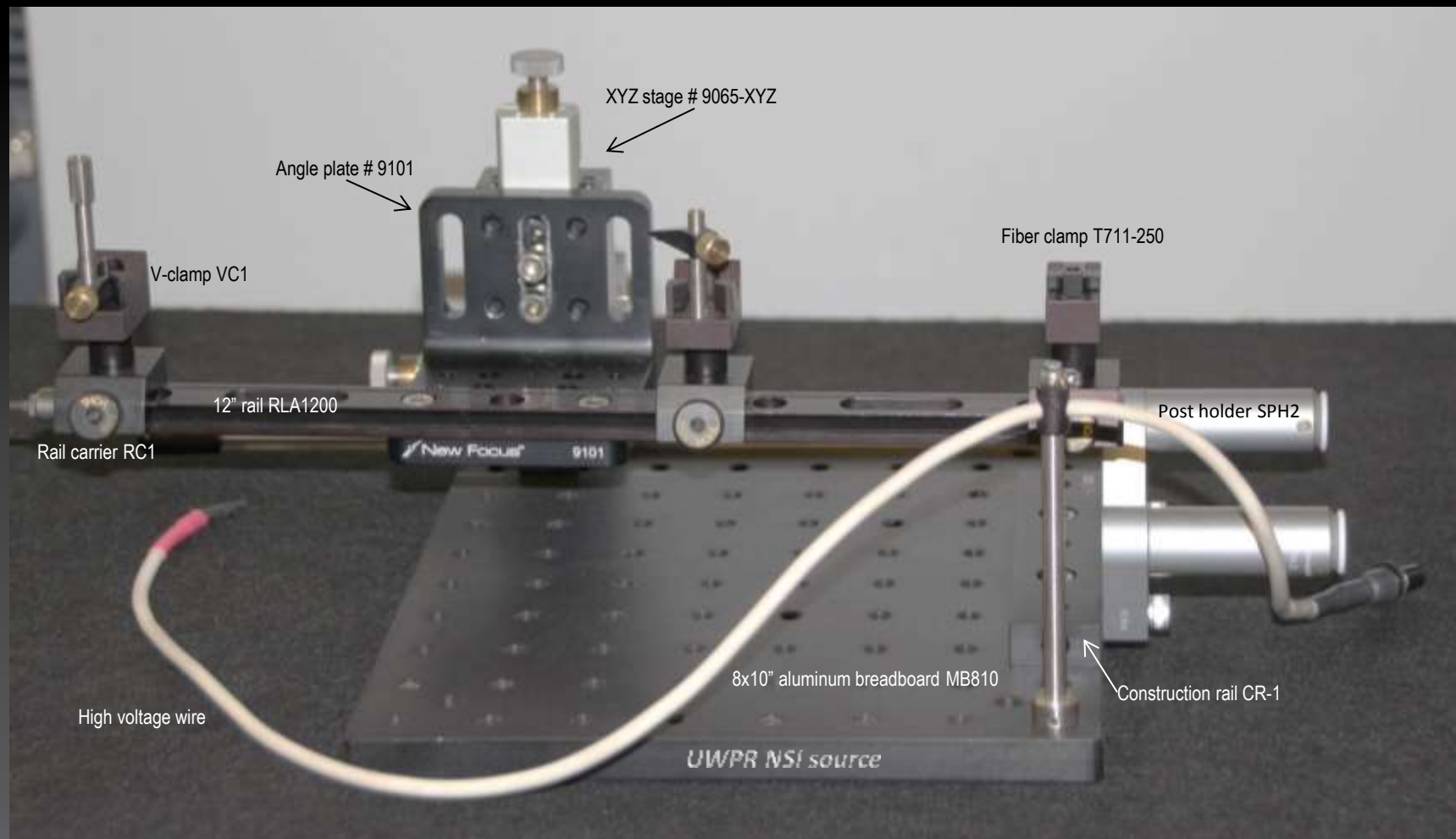
Nano-Spray-Ionization Source



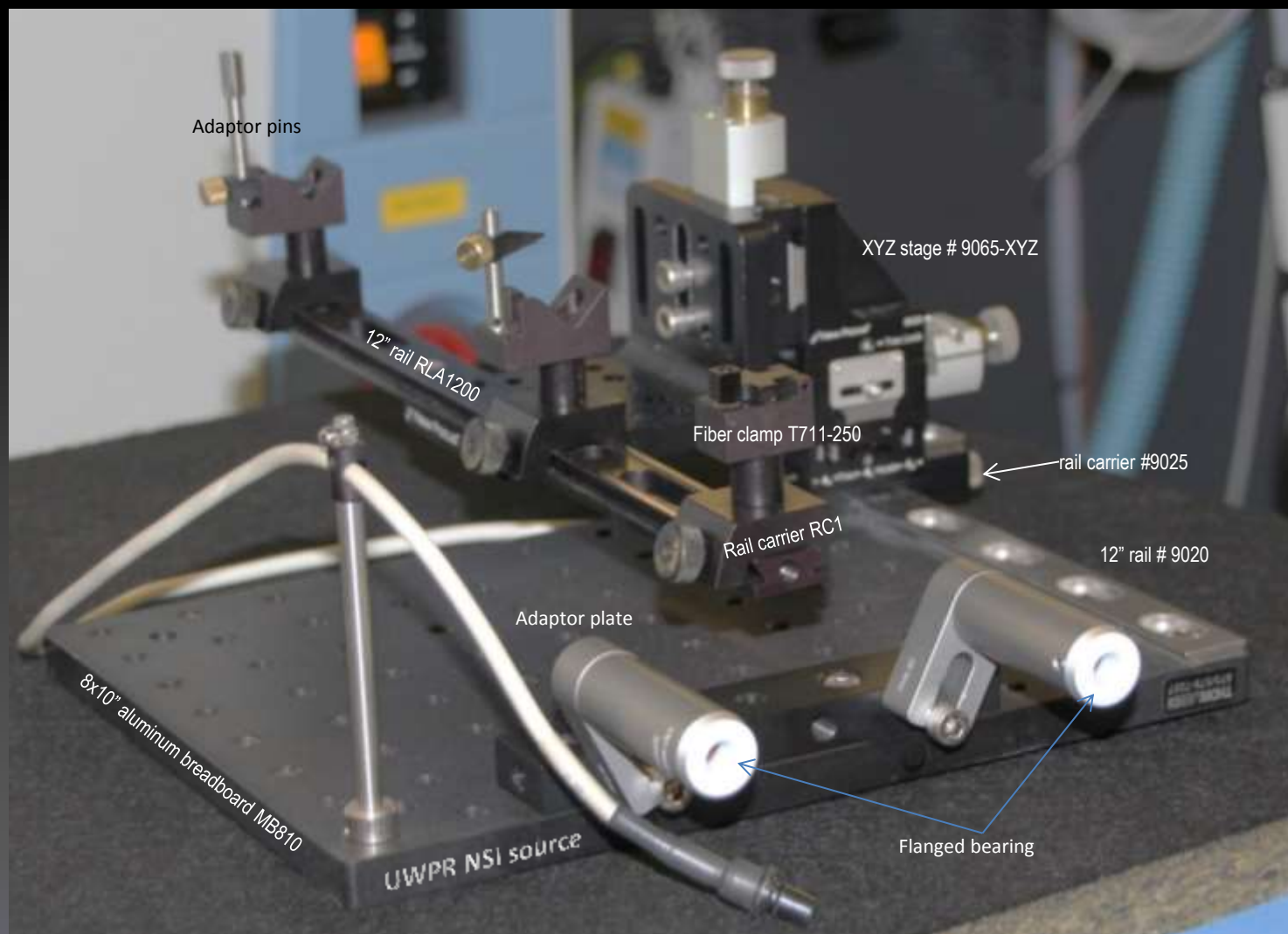
- The NSI source can be adjusted to fit all Thermo instruments (LTQ, Orbitrap, TSQ, Fusion)
- It offers all the benefits and features of commercial NSI sources:
 - Lower flow rates: no drying gas or thermal heating is required and increased sensitivity compared to higher flow ESI
 - Higher tolerance to a wide variety of liquid compositions than conventional ESI
- Additionally this NSI source offers significant improvements over most commercial NSI sources:
 - Great flexibility: it will not only accommodate in house packed columns of all length but also commercial columns, as well as configurations with traps, spray tips etc.
 - The source can be customized to your specific needs: e.g. mounting of cameras, LED lights or anything else you can think of.....
 - Additionally this NSI source can be used in conjunction with the Thermo NSI probes (static , dynamic and packed tip probe)
 - The open design allows for leaks to be detected immediately
 - Significantly lower cost
 - The NSI platform can also be used with the adaptor ring of the Thermo NSI source, eliminating the need to machine the adaptor bracket
 - All the parts are commercially available , except for the adaptor system, which is custom machined, e.g. by emachineshop.com
- The UWPR is currently using the NSI source on our Thermo instruments
- The plans for this NSI source are available for free, use at your own risk.... 😊
- This document and the xls file with all the part numbers should help you build your own source
- Note this is just a basic version, you can modify this source to fit your needs

UWPR NSI Source

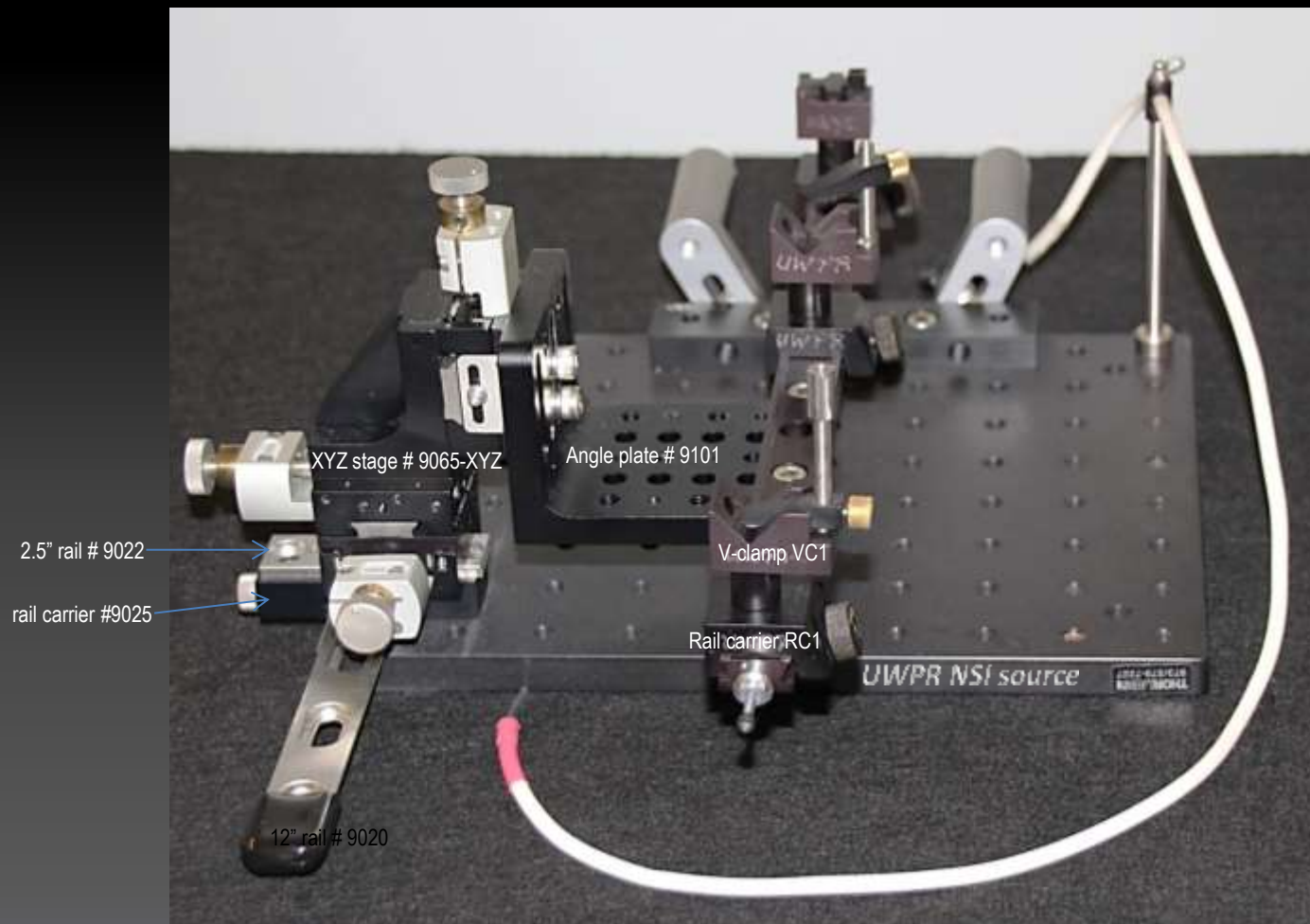
This is what we are going to build:



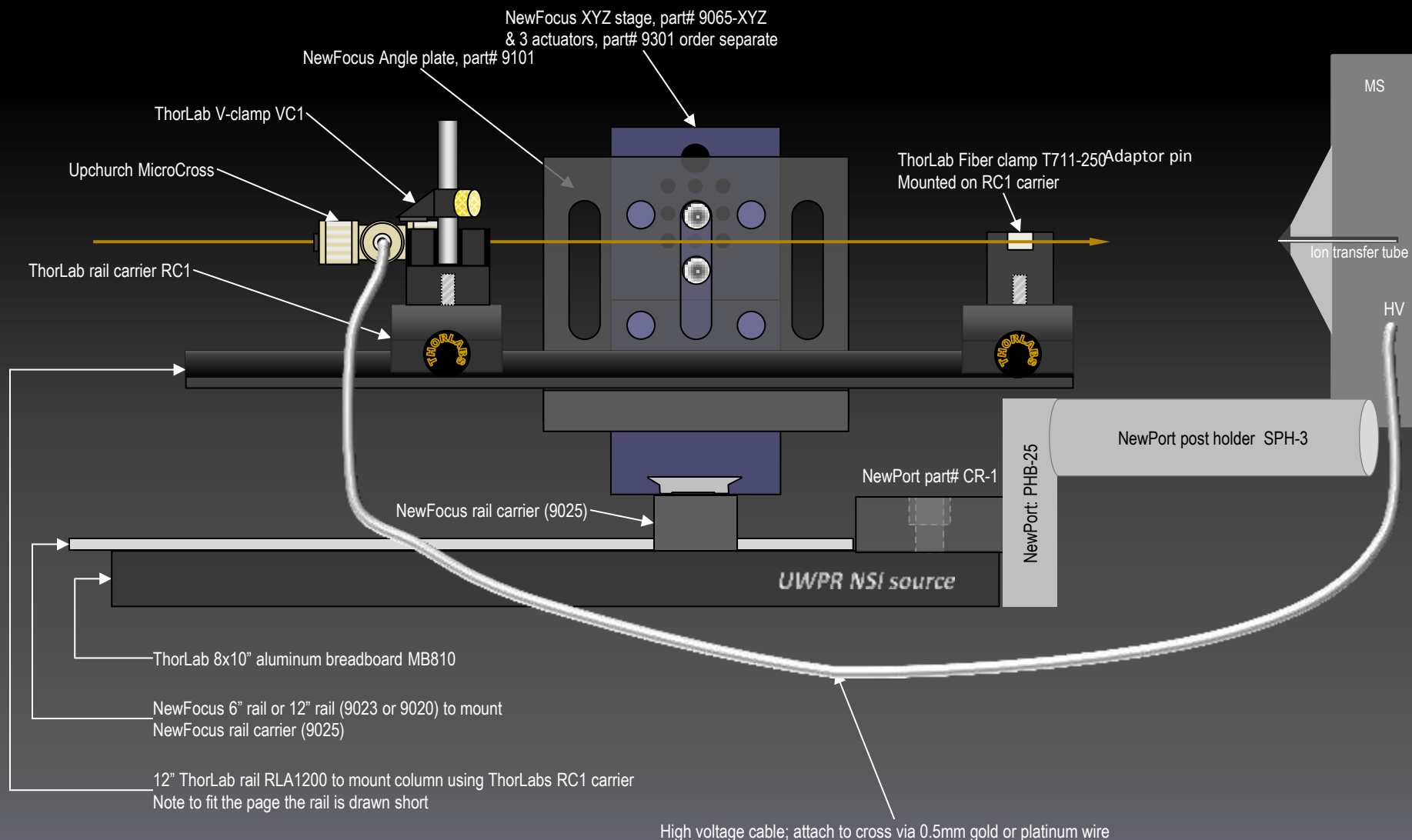
UWPR NSI Source



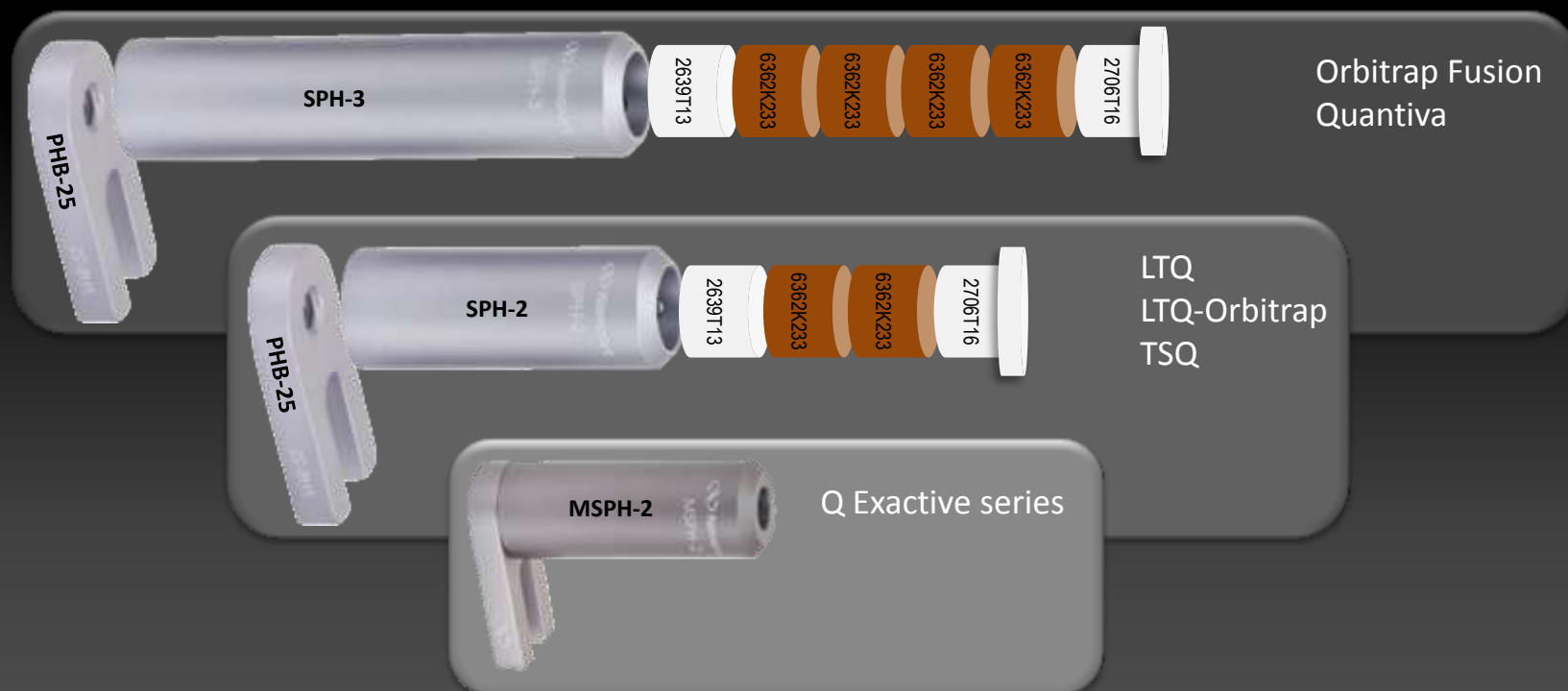
UWPR NSI Source



UWPR NSI Source



NSI Source: step 1



Note: you need two sets per instrument (one for each pin)

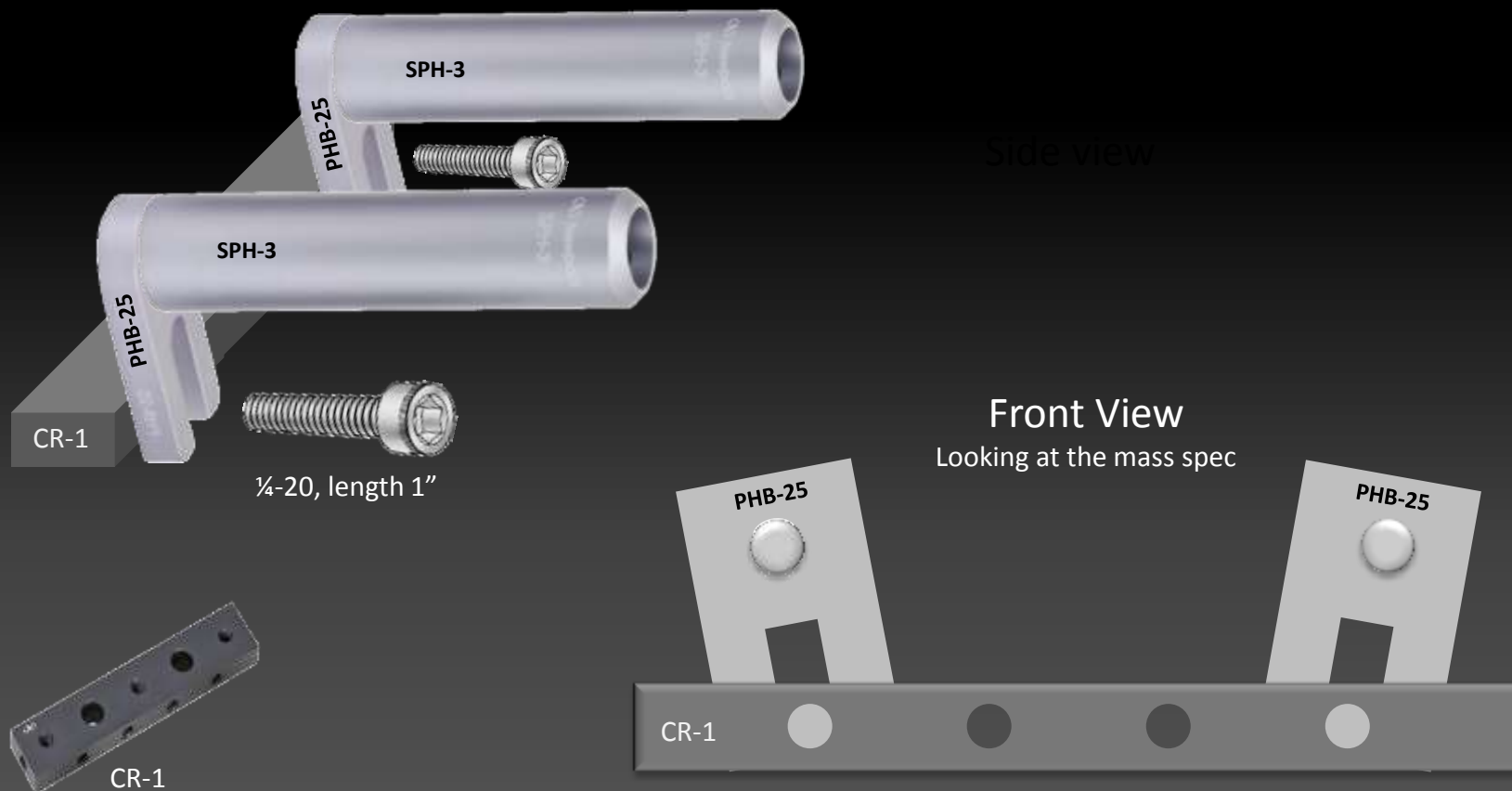
1. Attach base to post holder using the screw that came with the base
2. Slide the bushings into the post holder in the order shown above
3. Lightly tighten the set screw on top of the post holder to hold flanged bushing
4. Slide the assembled post holders over the pins on the instrument

SPH-3/SPH-2
PHB-25
2706T16
6362K233
2639T13
MSPH-2

Slim 1/2-in. Post Holder, 3-in. (76.2-mm) Ht. / 2-in.
Open Slotted Base, 1/4-20 Thread, 0.25-in Thick
PTFE Flanged Sleeve Bearing for 5/16" Shaft Diameter, 1/2" OD, 3/8" Overall Length
Rulon LR Sleeve Bearing for 5/16" Shaft Diameter, 1/2" OD, 1/2" Length
PTFE Sleeve Bearing for 3/8" Shaft Diameter, 1/2" OD, 1/2" Length
Miniature Optical Post Holder, 2.25 in., 1/4-20 Slotted Base

(Newport)
(Newport)
(McMasterCarr)
(McMasterCarr)
(McMasterCarr)
(Newport)

NSI Source: step 2



5. Slide the two post holders over the two pins on the mass spectrometer
6. While attached to the mass spec use two $\frac{1}{4}$ -20 screws to attach the construction rail to the two bases
Make sure the construction rail is level and tighten the screws

NSI Source: step 3

Use one #8-32 x 1/4" with a #8-32 washer and nut and one #8-32 x 3/8" with a nut, to attach 2.5" rail to rail carrier



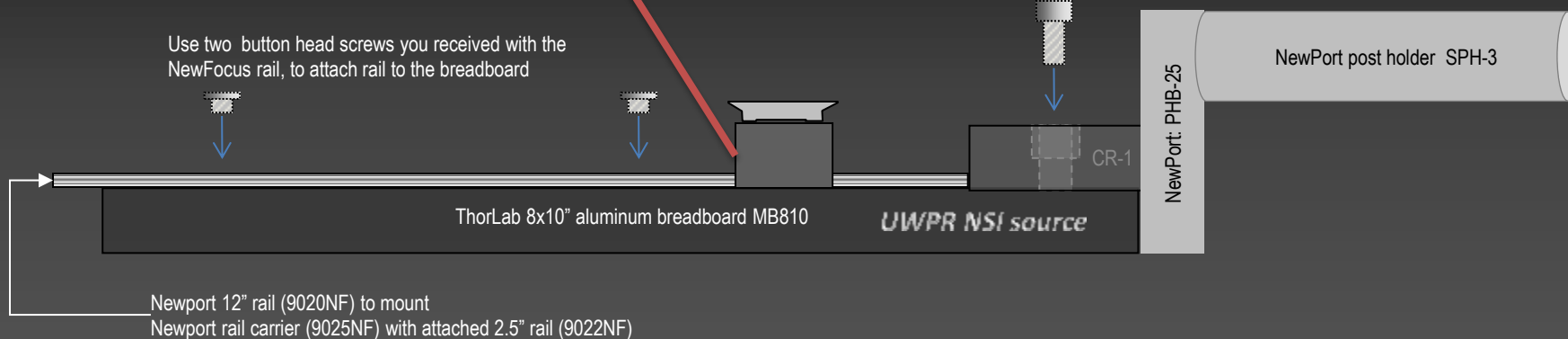
Newport 2.5" rail (9022NF)

Newport rail carrier (9025NF)

Replace #8-32 setscrew with Thumbscrew (91035A430)

Use two button head screws you received with the NewFocus rail, to attach rail to the breadboard

Use two 1/4"-20 x 3/4" screws to attach breadboard to CR-1



9022NF
9025NF
91035A430
9020NF
MB810

e-z-Trac™ Rail, 2.5 in. Length
e-z-Trac™ Post Carrier, 0.75 in. Width
18-8 Stainless Steel Knurled Head Captive Panel Screw
e-z-Trac™ Rail, 12.0 in. Length
Aluminum Breadboard, 8" x 10" x 1 1/2", 1/4"-20 Threaded

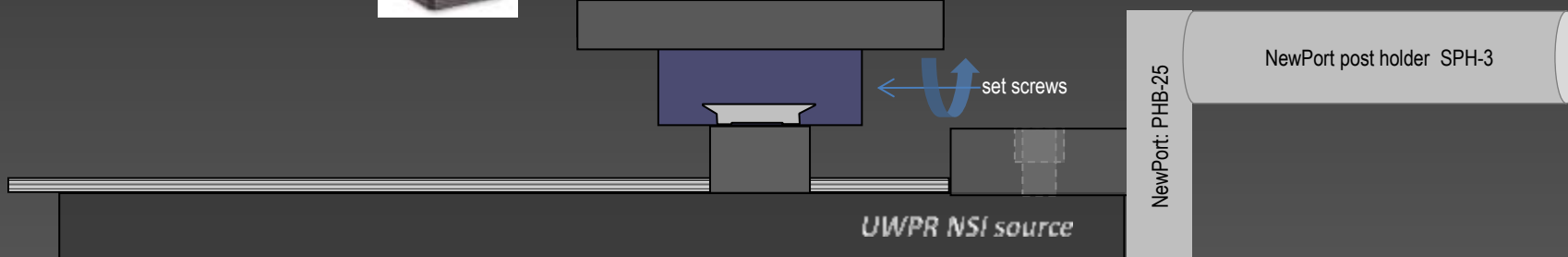
(Newport)
(Newport)
(McMasterCarr)
(Newport)
(Thorlabs)

NSI Source: step 4



Mount Newport XYZ stage, part# 9065-XYZ
To 2.5" rail using set screws provided with stage

Attach angle plate, part# 9101 to XYZ stage using
two 1/4"-20 x 3/8" screws

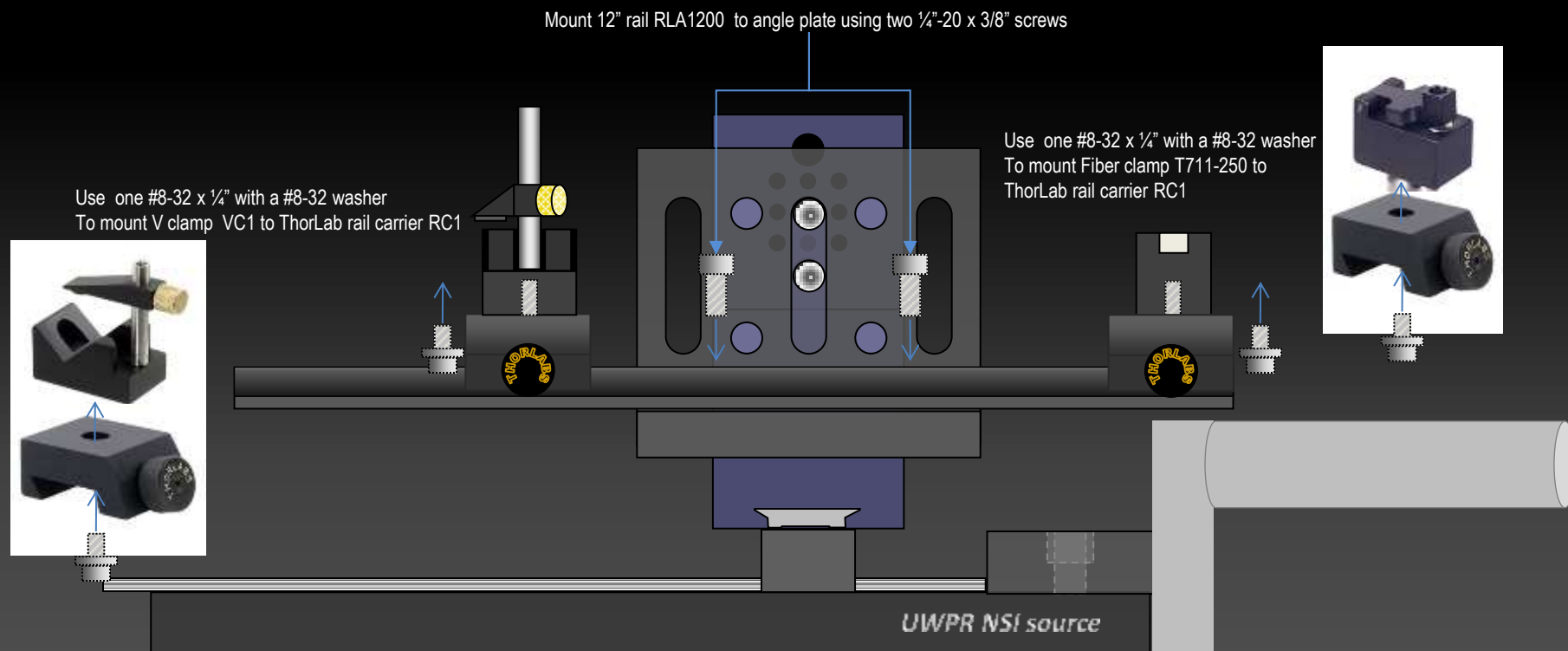


9065-XYZ
AJS100-0.5K-NL
9101NF

Pint-Sized XYZ-Translation Stage
High Precision Large Knob Adjustment Screw, 12.7 mm Travel, 100 TPI, No Lock
Angle Plate, 1/4-20 Thru Slots, 8-32 and 1/4-20 Threads

(Newport)
(Newport)
(Newport)

NSI Source: step 5



RLA1200
RC1
VC1
T711-250

Imperial Dovetail Optical Rail, 12"
Rail Carrier, Counterbored Hole 1"x 1"
Small V-Clamp with PM3 Clamping Arm
Post Mountable Fiber Clamp, 250mm

(Thorlabs)
(Thorlabs)
(Thorlabs)
(Thorlabs)



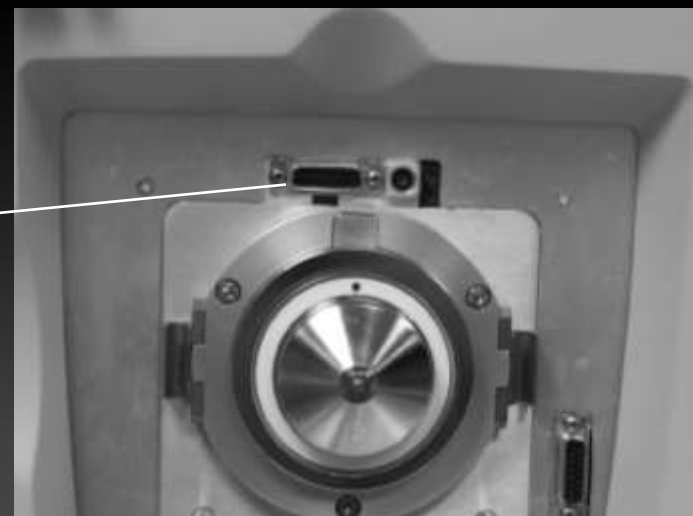
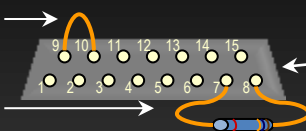
NSI Source: step 6a

To bypass interlock on ThermoElectron LTQ, TSQ, QE when using alternate nanospray source (NSI)

Note: Jumper and resistor can either be attached directly to the connector on the front of the LTQ or via a DB15 male connector (easier to change between different sources).

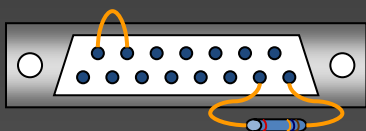
Jumper to bypass interlock between socket 9 and 10

10 K ohm resistor (2% var.) for NSI source recognition between socket 7 and 8



OR use DB15 male connector:

solder the jumper and resistor to the corresponding pins



DB15 male connector backside view



Solder Cup D-Sub Connector,
DB15 Male

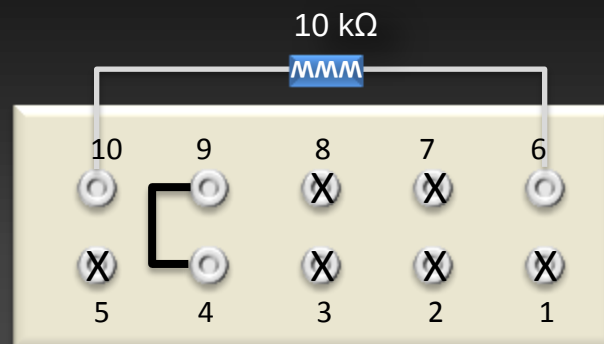
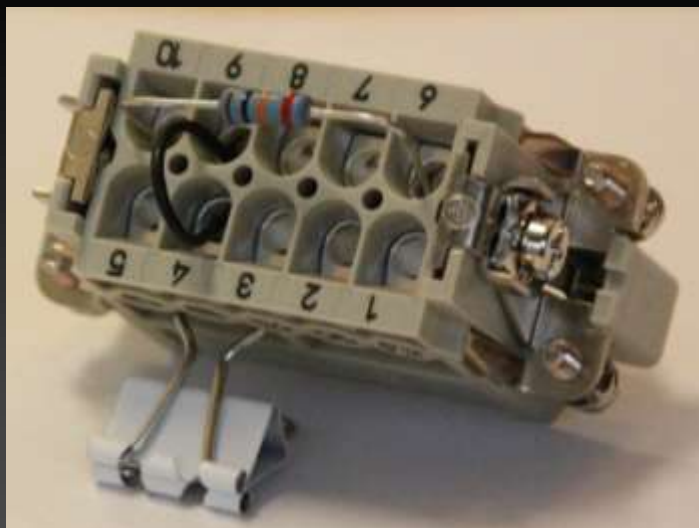
510-2002
510-2445
70201054

Emerson Network Power, D-Sub Plug; Thermoplastic; Plug; Copper Alloy; Solder Termination
Emerson Network Power, Hood; D-Sub; Chrome; Metalized Plastic; Steel; UL 94 VO; RoHS Compliant
Resistor; Metal Film; Res 10 Kilohms; Pwr-Rtg 0.25 W; Tol 2%; Axial; Epoxy

(Allied Electronics)
(Allied Electronics)
(Allied Electronics)

NSI Source: step 6b

To bypass interlock on ThermoElectron Fusion or Quantiva when using alternate nanospray source (NSI)



Male Plug HARTING 09200102612: Connect 10 kΩ resistor between pin 6 and 10 and jumper wire between pin 4 and 9
Optionally a ground wire can be attached to pin 7

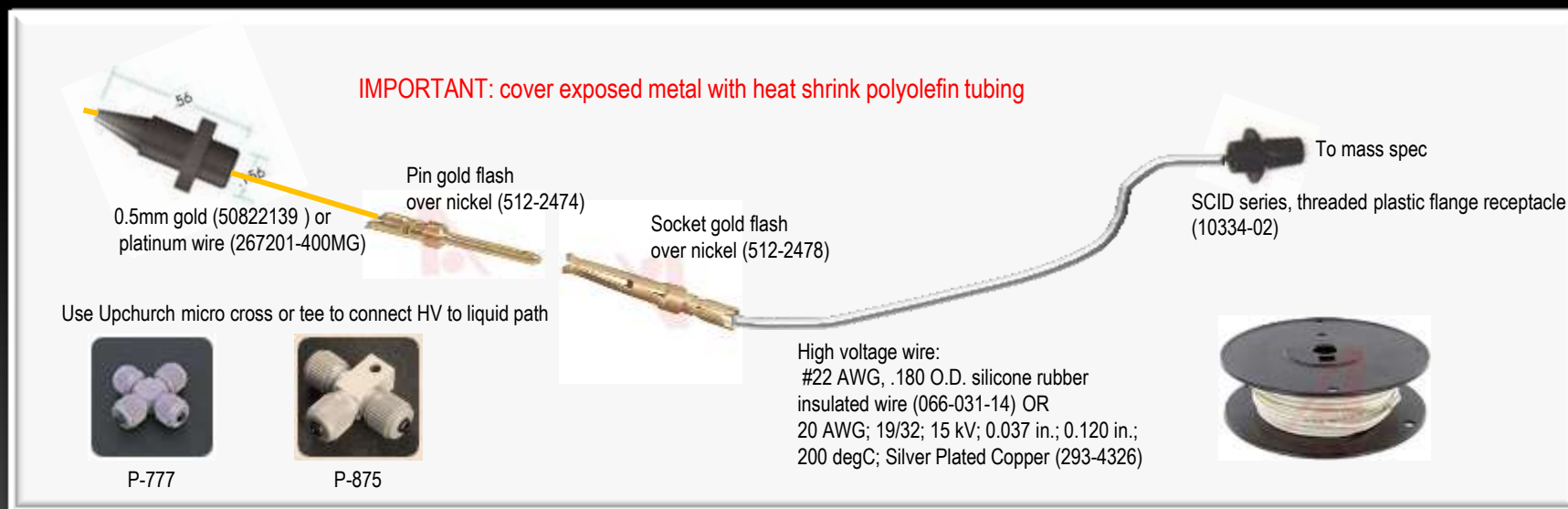
In case you wonder why I show it upside down... that's because looking at the instrument from the front, that is how it will plug into the instrument

70104585
70201054

Insert; 250 V; 16 A; Male; 10; RoHS Compliant; Han A Series; 10 A Product Size Mfr. Part#: 09200102612
Resistor; Metal Film; Res 10 Kilohms; Pwr-Rtg 0.25 W; Tol 2%; Axial; Epoxy

(Allied Electronics)
(Allied Electronics)

NSI Source: step 7



10334-02
Or
00004-95039

SCID series, threaded plastic flange receptacle assembly
Connector, Feed thru, 15KVdc, 7.5A, Connectronics

(Connectronics Corp)

(Unity Lab Services)

066-031-14
512-2474
512-2478
AA43288BU
P-777
P-875
F-172x
F-185x
P-116

High voltage wire, #22 AWG, .180 O.D. silicone /rubber insul.
20DF PIN CONT. LP
20DF SOCKET CONT. L.P.
Platinum wire; 99.95% ; Alfa Aesar; 0.5mm dia; 25cm; 4.21g/m
Microcross, .025" OD tubing sleeves, .006" THRU HOLE, PEEK
Microtee, w/ mount. whole, for .025" OD tubing sleeves, PEEK™
Ferrule, for .025" sleeves/P-416 NUT, PEEK™, BLACK (10 PK)
Tubing sleeve, 395µm (.015") ID x .025", PEEK™, GREEN (10 PK)
Plug, for micro fittings, PEEK™, BLACK

(Connectronics Corp)
(Allied Electronics)
(Allied Electronics)
(Fisher)
(Upchurch)
(Upchurch)
(Upchurch)
(Upchurch)
(Upchurch)

NSI Source: step 8

Note on the Fusion and Quantiva there is a inter-lock that needs to be pushed in
I used the miniature posts and a 0.5" 4-40 set screw



Adjust set screw to make sure the interlock is pressed in



MS3R



Set screw 4-40 0.5"



MSA25



ER90B

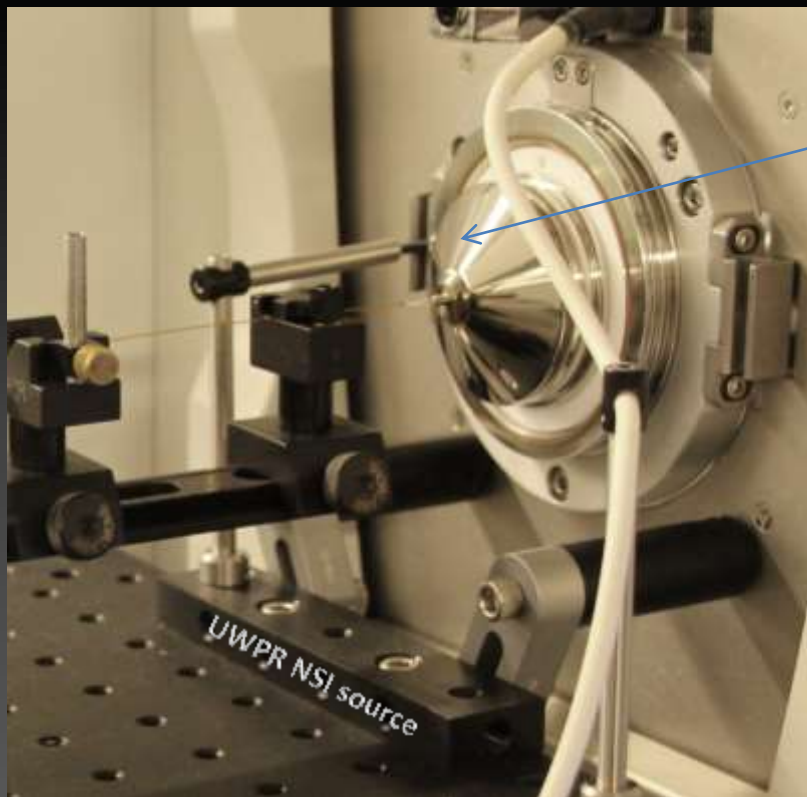
MS3R
MSA25
ER90B
Set screw 4-40 0.5"

Mini Series Mounting Posts, 6mm Diameter, 3" Long
Thread Adapter, 1/4"-20 to #4-40
Mini-Post Right Angle Adapter

(Thorlabs)
(Thorlabs)
(Thorlabs)

NSI Source: step 9

Note on the QE there is a inter-lock that needs to be pushed in
I used the miniature posts and a 1" 4-40 set screw



Adjust set screw to make sure the interlock is pressed in



MS3R

MS2R



Screw 4-40 1"



MSA25



ER90B

MS3R
MS2R
MSA25
ER90B
4-40 screw 1in long

Mini Series Mounting Posts, 6mm Diameter, 3" Long
Mini Series Mounting Posts, 6mm Diameter, 2" Long
Thread Adapter, 1/4"-20 to #4-40
Mini-Post Right Angle Adapter

(Thorlabs)
(Thorlabs)
(Thorlabs)
(Thorlabs)